

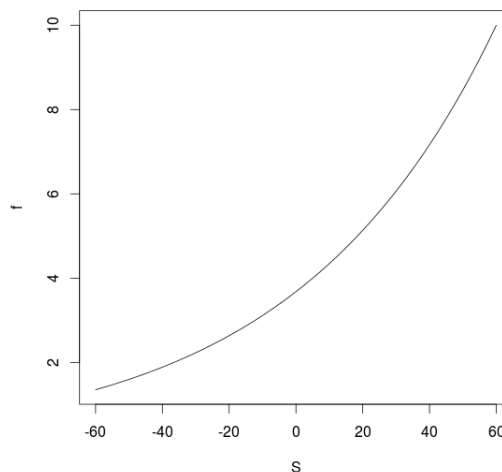
Machine Learning - MIRI Master (Final quiz - June 1, 2017)

Name:

Instructions:

- tick **clearly** the claims that you think are **true** with a $\checkmark$
- tick **clearly** the claims that you think are **false** with a $\times$
- if you want to “withdraw” an already ticked box, black it out as $\blacksquare$ (it will count now as unanswered)
- all questions are equally weighted (the headings define **blocks** of **ten** questions each)
- there is no obligation to answer individual questions, but at least half (**five**) questions in each block must be answered
- individual question grading: correct answers count +1 point, incorrect answers count -1 point; no answer counts 0 points (there are 60 questions = 60 points maximum)
- letting S be the number of points, the overall grade is obtained as

$$f(S) = 10 \exp\left(\frac{S}{60} - 1\right)$$



- deliver just **these sheets** back
- time: 2h

1. Complexity control and all that jazz.

- ☐ Based on training data alone, there is no means of choosing which model is better
- ☐ Complexity control is only necessary when data is high-dimensional
- ☐ The empirical error in the training set is always smaller (or equal) than the empirical error in the test set
- ☐ Using a larger training data set reduces the chances to obtain an overfitted model
- ☐ Regularization is intended to penalize models that are less complex than needed
- ☐ Cross-validation is mainly used for model selection purposes
- ☐ L_2 -regularization does not produce sparsity, as opposed to L_1 -regularization
- ☐ Reducing the hypothesis (model) space is a way of controlling complexity
- ☐ The VC dimension of a two-class linear classifier is a linear function of data dimension
- ☐ The VC dimension of a two-class classifier is always a finite integer

2. Bayesian classifiers.

- ☐ The Bayes formula transforms prior distributions into posterior distributions
- ☐ The numerator in Bayes formula is enough to perform classification, by taking simply the maximum over the classes
- ☐ The Bayes classifier is the best possible classifier when the classes are Gaussian
- ☐ The Bayes classifier is the best possible classifier when the true priors are known
- ☐ For normally distributed classes, Bayesian classifiers turn out to be quadratic discriminant functions
- ☐ For normally distributed classes, equal posterior probabilities yield linear discriminant functions
- ☐ The Naive-Bayes classifier does not make assumptions about data distribution for continuous variables
- ☐ The kNN classifier requires tuning of the number of neighbours, because we have a finite data sample
- ☐ $\sum_a P(a|b)P(b) = P(b)$, where A, B are discrete random variables
- ☐ $\sum_b P(a|b) = \sum_a P(b|a)$, where A, B are discrete random variables

3. Maximum Likelihood and GLMs.

- ☐ The likelihood is a function of the parameters for a given choice of data sample
- ☐ Logistic regression does not make assumptions about input data distribution
- ☐ Linear regression assumes normally distributed outputs, conditioned on the inputs
- ☐ In a GLM, the model tries to predict the expected value of the target using a linear function of the predictors and a suitable interface function
- ☐ We can obtain an error function as the negative log-likelihood of a problem
- ☐ The regression function is the best possible predictor, in the sense that it would achieve *zero bias*
- ☐ The regression function is the best possible predictor, in the sense that it would achieve *zero variance*
- ☐ The regression function is the best possible predictor, in the sense that it would achieve *zero noise*

- ❑ In statistics, bias and variance are related concepts: they represent the distribution of errors in the training and test sets, respectively
- ❑ The mean squared error is always preferred for GLM regression, because it is the only one that works in practice

4. Neural networks.

- ❑ An MLP needs no regularization, because backpropagation prevents arbitrary growth of the weights
- ❑ We can convert a non-linear model into a linear one by giving values to the non-linear adaptive parameters
- ❑ The backpropagation algorithm computes the partial derivatives of a given differentiable error function with respect to the network weights
- ❑ The backpropagation algorithm must be coupled with an optimization method (update rule) to make it a learning algorithm for a neural network
- ❑ The backpropagation algorithm is mainly used to compute the gradient vector of the error function at each step
- ❑ The nature of the target variable dictates the activation function for the output neurons
- ❑ The activation function for the hidden neurons could be a linear function to facilitate learning
- ❑ Both RBF and MLP neural networks can have one or more hidden layers of neurons
- ❑ An RBF neural network could in principle be trained with the backpropagation algorithm
- ❑ Regularization makes little sense in neural networks, because they are non-linear models

5. Kernels and SVMs.

- ❑ The kernel function defines kernel matrices whose elements are always non-negative
- ❑ Any positive linear combination of a number of kernel functions is a kernel function
- ❑ By choosing a valid kernel, we get an Euclidean distance in some Hilbert space
- ❑ In SVMs, the Lagrange coefficients α_n are positive for the support vectors and negative for the rest (the non support vectors)
- ❑ In order to “kernelize” a learning algorithm, this must be supervised to get meaningful targets
- ❑ The cost parameter (C) in a SVM has a role similar to that of a regularization parameter
- ❑ Increasing the value of C in a SVM (and everything else being equal), the margin cannot increase
- ❑ Increasing the value of C in a SVM (and everything else being equal), the number of support vectors can increase
- ❑ A positive semi-definite matrix may have negative elements in the main diagonal
- ❑ The VC dimension of a SVM depends on the kernel function it uses

6. Miscellaneous.

- ❑ The E-M algorithm refines a suboptimal solution obtained by k-means until a global optimum is found
- ❑ A Gaussian mixture model assumes that the data has been generated by a “big” Gaussian that can be decomposed as a finite mixture
- ❑ The k-means algorithm will discover the true clusters in the data, if given enough prototypes
- ❑ Bagging methods are based on the fact that, for unstable learners, variance can be greatly reduced with little or no increase in bias
- ❑ A Random Forest is “random” because decision trees are random learners (meaning that a single tree changes if we “execute” the algorithm again)

- ❑ A Random Forest is “random” because the *data* used in *each decision node* come from a different bootstrap resample
 - ❑ In Machine Learning, the lack of predictive variables can be compensated by more training data; in other words, there is no limit on the achievable predictive performance of a model, if we can gather enough data
 - ❑ We should optimize the number of folds in cross-validation, and separately for each modeling technique
 - ❑ In Machine Learning, better pre-processing can make a large impact on learning, and therefore on predictive performance
 - ❑ In a noiseless setting, at least theoretically speaking, there is no need for regularization
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